



Instructor: Date:

Learner: Regional Branch:

Program: Center:

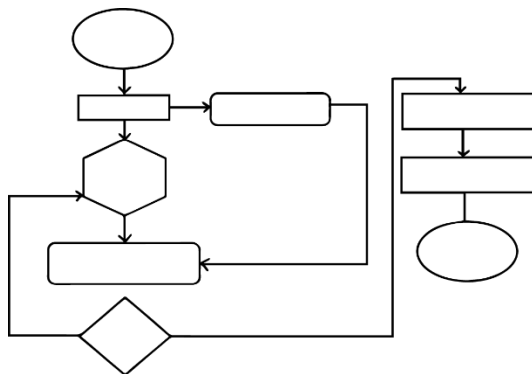
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Let's Start!

Let's study some other characteristics about DFD.



Learning Outcome:

Describe common activities at work using short sentences in familiar spoken and written situations.

Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Can Do Statements:

- I can explain the importance of direction of flow.
- I can talk about external entities.
- I can describe the parts related to data flow diagrams.

Ask your partner:

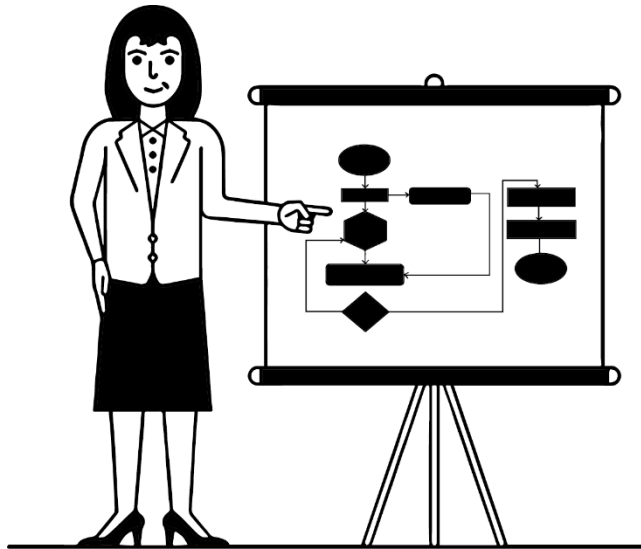
- Do you consider that flow direction varies in DFD? Why?
- Which can be the external entities in an environmental project or another project that you know?



Let's Explore!

A professor gives a lecture* about the Direction of Flow and the External Entities in a DFD.

*A "lecture" is a conference or speech you give to an audience.



Instructions: Read the text below.

To start, Direction of Flow represents the movement of information packets to or from different components that are illustrated by a hand arrow. In a diagram, you need to identify the arrows and names, which are both elements of a DFD. Remember that "DFD"

is the acronym for Data Flow Diagram, and they have some differences. If you analyze it,

you can identify that all parts have a name, but the flows don't have one because they represent the direction of flow.

Remember that a data flow has only one direction between symbols, but it can flow in both directions between a process and a data store. Additionally, if you don't make the difference between single arrows and two separate arrows with specific directions, it can be difficult to identify an update, read or write actions.

For example, when a single arrow points towards the process, it depicts a 'read' action; in this case, the process does not alter the content of the store. Finally, an update contains both read and write actions, and you can have 2 arrows on both directions.

On the other hand, an external entity can represent a person and an organization, both of them are outside the system, and they receive or send data from the system. if you don't want to get confused with the other parts, you can identify external entities with an

alpha character. Also, you have to write both an identifier and the name of the entity. Finally, when the external entity supplies data to the system, it represents a source As well, when it receives the data, it's a sink. Anyway, both the sources and the sinks are external entities and parts of the DFD.





Let's Practice!

The data flow moves in, out and around a system.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: and – both, both parts, and the sinks, it is both, both - and the sink.

1. I will talk about directions of flow _____ external entities which are _____ elements of a DFD.
2. Anyway _____, the sources _____ are external entities.
3. When the external entity supplies data to the system, _____ a source _____.



Let's Practice!

Data flow diagrams documents processes in a system.

Instructions: Select the correct option to correctly complete each sentence.

1. If you don't have arrows, (it's not possible / it's not possibly / it's don't possible) to identify the direction of the flow.
2. If you analyze a DFD, you (can't forget / can identify / can't identify / identify) can forget / can't forgets) that all parts have a name.
3. If you (no make / don't make / don't made) the difference between single arrows and two separate arrows, it's difficult to identify updates, reads and writes.





Extension Activity

The Directions of Flow and External Entities.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Read information about the parts related to data flow diagrams.
- Create a poster in which you can describe and exemplify these parts.
- Prepare an oral presentation using the zero conditional and conjunctions to explain the topic.
- Share your work to your classmates and check pronunciation with your instructor.

Self-Assessment

Yes

No

- I explain the importance of direction of flow. ☐ ☐
- I talk about external entities. ☐ ☐
- I describe the parts related to data flow diagrams. ☐ ☐



Target Words

- **packets** (noun): a part of a data or a small unit
e.g. Direction of Flow represents the movement of information packets to or from different components.
- **update** (noun): new, old or current information which has movement in a data flow diagram
e.g. An update contains both read and write actions, and you can have 2 arrows on both directions.
- **single** (adjective): just one part or section
e.g. if you don't make the difference between single arrows and two separate arrows with specific directions, it can be difficult to identify an update.
- **towards** (preposition): in direction of something
e.g. When a single arrow points towards the process, it depicts a 'read' action;
- **alter** (verb): to change or modify
e.g. In this case, the process does not alter the content of the store.
- **get confused** (expression): to not understand something
e.g. If you don't want to get confused with the other parts, you can identify external entities with an alpha character.



Learning Points

Zero Conditional With The If Sentence In Negative.

You can use zero conditional in both negative and affirmative statements to talk about a situation that is a real fact.

You use this conditional to explain two situations that complement each other and that are always real. The first part of the conditional is called the main clause, this explains the condition. The second one explains the result, consequence or direct relation main clause. When the main clause is a negative statement it shows a negative condition.

Remember to use the present simple form in negative after the "if" sentence.

The structure is like this:

If + Present Simple (negative form) + comma (,) + Present Simple

Example:

- *You do not have to worry about the direction of the flow **if** you have a basic diagram.*
- ***If** you **don't** analyze a Data Flow Diagram correctly, you don't recognize all the parts.*

Remember that you separate the two clauses by a comma (,) When you start with the if clause.

Examples

- *If you analyze it you can identify that all parts have a name but the flows don't have one, as they just represent the direction of flow.*
- *If you don't make the difference between them; it's difficult to identify an update, read or write actions.*
- *If you don't want to get confused with the other parts, you can identify external entities with an alpha character.*





Learning Points

Conjunctions, Both... And.

Conjunctions are linking words that you can use to connect words or sentences. **Both ... and** are linking expressions.

You can find different types of conjunctions, they can be coordinating, correlative or subordinating conjunctions. You use "**and**", which is a coordinating conjunction that means in addition to, when you want to connect words, phrases, or independent clauses.

Example:

- *She wants to do a data flow diagram **and** a manual.*
- *He is checking the DFD **and** she is working on the manual.*

You can use "**both**", which is a correlative conjunction, when you want to express that two things are included in a phrase. Additionally, in the same sentence you can use **both ... and** when you want to mention the connection between two ideas.

Example:

- ***Both** squares and triangles are shapes.*
- *I bought **both** a software **and** a new application.*

Both	Use	Example
Both with pronouns	You can use both after a subject pronoun or an object pronoun	<i>They both buy new programs.</i>
Both with nouns	You can use both before a noun:	<i>Both programmers are good.</i>
Both as a pronoun	Both can be its own pronoun.	<i>The system engineers are in the house. Both are developing a new software.</i>

Examples

- *Also, you have to write both an identifier and the name of the entity.*
- *Anyway both, the sources and the sinks are external entities and parts of the DFD.*